

Total No. of Questions : 8]

SEAT No. :

PD4667

[Total No. of Pages : 2

[6404]-173

B.E. (Information Technology)

WIRELESS COMMUNICATIONS

(2019 Pattern) (Semester - VII) (414445 D) (Elective - IV)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume Suitable data if necessary.

Q1) a) What is OFDM technique? Also, explain OFDMA transmitter and receiver. [9]

b) Explain the Principal of TDMA. What are different features of TDMA? [9]

OR

Q2) a) What is MIMO? Explain two formats of MIMO. [9]

b) How Code Division Multiple Access Technique is implanted while accessing a channel for multiple users? Support your theory answer with example. [9]

Q3) a) What are the different challenges in WAP? Also, write down the advantages and disadvantages of WAP. [9]

b) Explain SigFox protocol in details. [8]

OR

Q4) a) What is NFC? What are the different characteristics of NFC? [9]

b) What is Li-Fi? Explain the working principle of Li-Fi technology. [8]

Q5) a) What is post-quantum cryptography? Explain in details. [9]

b) What is security? What are the different security issues in 1G, 2G, 3G, and 4G? [9]

OR

P.T.O.

- Q6)** a) Explain Bluetooth Security Architecture in detail. [9]
b) What are Wireless Security tools? Explain URH and Kismet in details. [9]

- Q7)** a) What is quantum Technology? Explain quantum Technology for a 5G/6G wireless network. [9]
b) State and explain various applications of wireless technology. [8]

OR

- Q8)** a) Explain how 5G NR (New Radio) working along with its benefits. [9]
b) Explain Simultaneous Transmission and Reflection (STAR) for 360° coverage in details. [8]

